Elena Rostova

Senior Distributed Systems & Storage Engine Engineer

Specialist in Asynchronous Rust, Raft Consensus, LSM-Tree Internals & io_uring

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PROFILE

Distributed systems engineer with 8+ years specializing in high-throughput transactional storage engines, write-ahead logging (WAL), and LSM-tree architectures. Deep expertise in asynchronous Rust (Tokio), Raft consensus protocols, lock-free concurrency, and zero-copy Linux asynchronous I/O using io_uring.

EXPERIENCE

Principal Storage Engineer 03/2021 - Present

ByteScale Systems - Stockholm

Architected distributed key-value storage engine in Rust with multi-Raft partition groups. Integrated Linux io_uring for non-blocking disk writes, sustaining 2.5 million sustained IOPS per storage node with p99 tail latency under 0.8ms.

Stack: Rust, Tokio, io_uring, Raft, RocksDB, gRPC, Protobuf, eBPF

Senior Distributed Systems Engineer 06/2018 - 02/2021

CloudData Labs - Stockholm

Engineered multi-group Raft partition coordinator for distributed database. Optimized LSM compaction pipelines and block cache eviction policies, reducing write amplification factor from 18x to 5.2x.

Stack: Rust, C++, LSM-Trees, Write-Ahead Logging, Paxos, Zookeeper

Systems Programmer 09/2016 - 05/2018

Nordic Infrastructure - Gothenburg

Authored high-performance network file storage servers in C and Rust with custom slab memory allocators and zero-copy sendfile transfers.

Stack: C99, Rust, POSIX Sockets, Epoll, Linux Kernel, GDB

Database Intern 06/2015 - 08/2016

Ericsson Research - Kista

Benchmarked lock-free B-tree concurrent indexing algorithms under high write contention workloads in C++.

Stack: C++, Multithreading, Atomics, Valgrind, Perf

CORE SKILLS

Distributed Systems

[Raft Consensus] [Paxos] [Multi-Raft]
[Distributed Locking] [Failure Detectors]
[Split-Brain Prevention]

Storage & Linux I/O

[LSM-Trees] [Write-Ahead Logging] [io_uring]
[Direct I/O] [mmap] [eBPF] [RocksDB] [SSTables]

Languages & Frameworks

[Rust] [Tokio] [C (C99)] [C++17/20] [gRPC]
[Protobuf] [Flamegraphs]

Databases & Tooling

[PostgreSQL] [Redis] [DuckDB] [Linux Kernel]
[Docker] [Cargo] [GDB] [Perf]

SYSTEMS ARCHITECTURE & PROJECTS

raft-rs Distributed Consensus

Pure-Rust implementation of Raft with pipelined leader append entries and dynamic cluster membership.

lsm-uring Storage Engine

Embedded LSM-tree database engine utilizing io_uring submission queues for NVMe direct I/O.

EDUCATION

M.Sc. Distributed Computing

KTH Royal Institute of Technology

LANGUAGES

Swedish	Native
English	Fluent / C2
German	Professional